

How the small investor can beat the market

By buying stocks that are selling below their liquidation value.

Joel M. Greenblatt, Richard Pzena, and Bruce L. Newberg

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Past experience and common sense tell us that the individual investor has little chance of outperforming the stock market. After all, even the billion dollar institutional funds, with their greater access to timely information and comprehensive analysis, have been notably unsuccessful in their efforts to achieve superior long term investment results.

Ironically, it is the expertise of Wall Street's portfolio managers and investment analysts that serves as the explanation for the disappointing performance of their equity funds and stock recommendations. Competition between these investment professionals, according to current academic theory, assures that stocks remain efficiently valued and that all publicly available information is reflected in their prices. Since this theory concludes that bargain purchases are therefore impossible, academics argue that attempts to consistently outperform the market are futile.

We should recall, however, that Wall Street research houses limit their coverage to fewer than 500 actively traded issues (Merrill Lynch being the exception with approximately 1100 stocks closely studied). Meanwhile, the NYSE trades 2000 stocks, the Amex trades 1000 companies, and the OTC market trades another 7000 issues that are required to provide relatively full disclosure to the S.E.C. Under these circumstances, the individual may in fact be able to locate unrecognized values in the nearly 9000-stock second tier not closely followed by the "experts."

In an effort to discover whether inefficiently priced, undervalued securities do exist, we turned to the acknowledged father of security analysis, the late Benjamin Graham. In his textbook, *Security Analysis*,

he outlines in little more than a page the opportunities to be found in stocks selling below their liquidation value. In studies between 1923 and 1957, Graham reported superior results when market levels enabled him to buy a diversified list of these bargain stocks. We had followed for several years the dramatic success of companies that the market priced below their value in liquidation, and so we decided to update Graham's studies to see if his simple fundamental approach still provided returns that could not be explained by an efficient market.

METHODOLOGY

As an underlying assumption in our work, we used Graham's rough calculation of liquidation value:

Current Assets (cash, accounts receivable, inventory, etc.),

less

Current Liabilities (short term debt, accounts payable, etc.),

less

Long Term Liabilities (long term debt, capitalized leases, etc.),

less

Preferred Stock (claim on corporate assets before common stock),

divided by

Number of shares outstanding,

equals

Liquidating Value Per Share.

The assumption that current assets less all liabilities and preferred stock approximates liquidation value (excluding fixed assets entirely) is based on

the idea that any loss on the liquidation of inventories and receivables will be made up through the liquidation of fixed assets (land, plant, machinery). This is of course only a rough screening device to sort out the most likely prospects for undervaluation out of the thousands available.

Clearly, on a theoretical level, it is possible for a stock to be selling below liquidation value if the discounted flow of its future earnings results in such a lower valuation and if there are no immediate prospects for liquidation. Firms losing money, for example, or those firms expected to lose money in the future, might reasonably be selling at sub-liquidation prices. In an attempt to minimize this possibility for the purposes of our study, we did not consider stocks that had shown a loss over the preceding 12 months. Graham found that, by eliminating those stocks reporting losses, he could still form diversified bargain lists during many market periods.

In designing our computer simulations to test Graham's statistical criteria, we selected 15 segments of 4 months each over a six-year period in which the over-the-counter (NASDAQ) averages halved and then doubled. (Approximately 60% of our selected stocks were traded in the OTC market.) The period under study ran from April 1972 to April 1978; it included the great market plunge of late 1974 and the subsequent period's strong recovery. The obvious advantage of such a volatile period is that we were able to observe the performance of our selected stocks during extreme market conditions.

We attempted to select a statistically significant and unbiased sample of stocks by compiling data on all firms listed in the Standard & Poor's stock guide with market values over \$3 million beginning with the letter A or B. This sample represented approximately 15% of all stocks listed in the guide, or 750 companies.

The buying rules we employed were varied, so that we could observe the effects of changing our statistical requirements for purchases on the performance of our simulated portfolio. We looked exclusively at three factors: (1) price in relation to liquidation value, (2) price/earnings ratio, and (3) dividend yield. The computer bought any stock that met our chosen criteria without any further analysis. We sold a stock after a 100% gain or after 2 years, whichever came first.

We calculated our returns by assuming an equal dollar-weighted amount invested in each stock in the portfolio. Therefore, the percentage gain for the entire portfolio was merely an average of the percentage gain of the individual stocks. Although the NASDAQ index we used for comparative purposes was computed on the basis of change in total market value (larger capitalization stocks carry more weight in this

index), we also included the equally-weighted Value Line index with our results to aid in the comparison.

INVESTMENT RESULTS

The results we achieved appear in Table I and are explained in detail here.¹

Portfolio 1:

Price/liquidation value: ≤ 1.0 ;

P/E: floating with bond yields;

Dividends: no dividend requirements.

In the first screen of our stock universe, we bought stocks selling below their liquidation value per share, below a floating P/E constraint related to the prevailing triple A bond yield, and required no minimum dividend yield. We required a P/E corresponding to twice the prevailing triple A yield in each period (e.g. triple A yield = 8%; required P/E equal or below the reciprocal of 16%, or 6.25).

During the 15 4-month periods our constraints dictated a position in the market, we averaged an annual compounded rate of return of 20.0% before dividends, commissions and taxes. The OTC index appreciated at an annual compounded rate of 1.3% during this same period.

We would expect higher returns to accrue to riskier investments to compensate us for taking on the additional risk. Therefore, we also studied the volatility of the returns of our selected stocks with that of the NASDAQ market average. (During this period, the NASDAQ averages significantly outperformed the S&P indexes of larger companies.) A regression of our Portfolio 1 return and the OTC market return over the 15 periods resulted in the following:

Portfolio 1 return = $+6.14 + .836$ (NASDAQ return),
(4 month period)

Portfolio 1 standard deviation = 14.15;

OTC portfolio standard deviation = 12.75.

If we assume that funds devoted to investment in securities meeting our constraints constitute only a portion of a diversified portfolio, then the beta contribution of Portfolio 1 would be .836. Our Portfolio 1 is undiversified, however, so this low beta coefficient could be attributed simply to a low covariance with the market. We calculated the beta that the portfolio taken by itself would achieve if perfectly diversified by dividing our undiversified beta by the correlation coefficient. Therefore, our adjusted beta for a perfectly diversified portfolio would be .836/.753, or 1.11.

The question of variation of return, though, is academic if this area of the market is, as it appears,

1. Footnotes appear at the end of the article.

TABLE I
Returns on Four Selected Portfolios
and Number of Stocks in Each Portfolio in Each Period
(4-month periods, April 1972 – April 1978)

Period	OTC Return	Value Line Return**	Portfolio 1 (1, fltg. P/E, 0)	Portfolio 2 (.85, fltg. P/E, 0)	Portfolio 3 (1, 5, 0)	Portfolio 4 (.85, 5, 0)
4/72-8/72	-1.1%	-8.0%	- / 0	- / 0	- / 0	- / 0
8/72-12/72	+2.9%	+3.7%	- / 0	- / 0	- / 0	- / 0
12/72-4/73	-19.4%	-15.4%	- / 0	- / 0	- / 0	- / 0
4/73-8/73	-2.8%	-9.2%	-10.9%/1	-10.9%/1	- / 0	- / 0
8/73-12/73	-12.1%	-15.8%	+2.4%/8	+11.5%/5	+21.1%/4	+28.1%/3
12/73-4/74	-5.8%	+4.5%	+8.6%/15	+9.5%/9	+9.9%/9	+15.2%/6
4/74-8/74	-28.2%	-22.1%	-18.1%/16	-22.7%/8	-19.2%/10	-22.4%/5
8/74-12/74	-4.1%	-13.7%	-5.8%/29	-4.7%/11	-4.3%/21	-0.6%/8
12/74-4/75	+31.3%	+28.5%	+30.1%/46	+26.4%/22	+31.7%/38	+28.5%/19
4/75-8/75	+1.0%	+10.2%	+19.7%/52	+24.0%/27	+19.8%/44	+24.8%/24
8/75-12/75	-1.8%	-4.5%	+2.6%/51	+12.7%/26	+5.2%/44	+14.2%/24
12/75-4/76	+16.1%	+23.8%	+34.4%/43	+29.6%/25	+35.0%/39	+29.6%/24
4/76-8/76	-0.4%	-1.0%	-4.1%/32	+0.8%/17	-2.7%/30	+0.8%/17
8/76-12/76	+9.1%	+0.0%	-1.0%/26	-1.0%/14	-1.0%/24	-1.0%/14
12/76-4/77	-2.5%	+5.0%	+16.6%/21	+25.3%/12	+16.6%/21	+25.3%/12
4/77-8/77	+4.9%	+3.4%	+4.4%/17	+4.4%/10	+4.4%/17	+4.4%/10
8/77-12/77	+5.4%	+0.7%	+12.0%/16	+13.9%/8	+13.9%/6	+19.1%/6
12/77-4/78	+9.2%	+1.0%	+16.5%/20	+21.4%/9	+17.8%/7	+23.1%/7
Annual Compound						
Return						
Before Tax & Comm.	+1.3%*	-0.3%	+20.0%	+27.1%	+32.2%	+42.2%
After Tax & Comm.			+11.3%	+16.5%	+20.1%	+29.2%

NOTES ON TABLE I

Example:

(1; 8; 0) Price/liquidation value equal to or less than 1.0; P/E less than or equal to 8;
Dividend yield equal to or above 0%.

+10.3/21 4-month percentage increase in portfolio equal to 10.3%;
21 stocks held in portfolio at beginning of period.

* 2.0% for port. 3 & 4

** equally dollar weighted

inefficient. Given a choice between an expected return of +1.3% with a standard deviation of 13%, or a return of 20% with a standard deviation of 14%, a prudent investor would opt for the latter without too much reservation about the marginally greater variability in return.

Our data use only 15% of the S&P stock universe, so there appears to be an opportunity to obtain a diversified portfolio of between 50 and 350 stocks that meet the constraints of Portfolio 1. In following our constraints, incidentally, we were unable to find any "bargains" in the market peak period of April 1972 to April 1973.

For the purposes of calculating a return after taxes and commissions, we assumed commissions of 2.5% on purchase price plus a 2.5% bid/ask spread (the bid/ask spread was applied to the 60% of our stocks that were purchased over-the-counter), a 2.5% commission on selling price, and a 25% capital gains

tax (over 90% of the stocks were held long enough to qualify for capital gains treatment). Given these assumptions, our return averaged to a compounded annual rate of 11.3% before dividends. Dividends averaged between 3% and 4% annually.

Portfolio 2:

Price/liquidation value: ≤ 0.85 ;

P/E: floating with bond yields;

Dividends: no dividend requirements.

After we limited the purchases in Portfolio 1 to stocks selling below 85% of liquidation value, the returns increased to a 27.1% annualized rate before dividends, commissions, and taxes (compared with the market's 1.3% annual performance). After taxes and commissions, this return approximated 16.5% annually. The regression worked out to:

Portfolio 2 return = $+8.44 + .752(\text{NASDAQ return})$
(4 month period);

Portfolio 2 standard deviation = 14.58;
 OTC Portfolio standard deviation = 12.75;
 Adjusted beta = 1.14.

Portfolio 3:

Price/liquidation value ≤ 1.0 ;
 P/E: ≤ 5.0 ;
 Dividends: no dividend requirements.

When we used a low constant P/E ratio coupled with a discount to liquidation value, our returns were significantly improved to a 32.2% annualized rate before dividends, commissions, and taxes. After taxes and commissions, our return falls to 20.1% per year, compared to the OTC market's return of 2.0% during the 14 periods we had a position in the market.

No stocks were purchased until August 1973 using the parameter of a P/E below 5. The portfolio also entered the market closer to the trough and with more conservatively valued stocks. We outperformed the OTC index by 5% or more in 9 four-month periods, while we underperformed the market by 5% in only one period. The regression analysis was:

Portfolio 3 return = $+9.9 + .753(\text{NASDAQ return})$;
 (4 month period)

Portfolio 3 standard deviation = 14.35;
 OTC portfolio standard deviation = 13.16;
 Adjusted beta = 1.09.

Portfolio 4:

Price/liquidation value ≤ 0.85 ;
 P/E: ≤ 5.0 ;
 Dividends: no dividend restrictions.

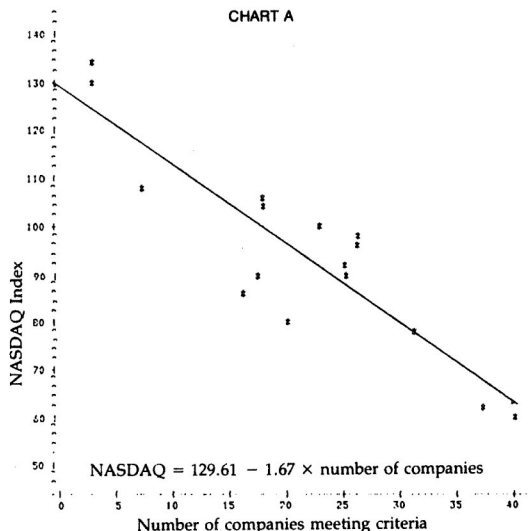
Our most successful screen. It resulted in an annualized return of over 42.2% before dividends, commissions and taxes. The result before dividend returns approximated 29.2% for the 14 periods studied, compared to the 2.0% annual return of the OTC market. The regression analysis was:

Portfolio 4 return = $+12.83 + .671(\text{NASDAQ return})$;
 (4 month period)

Portfolio 4 standard deviation = 14.94;
 OTC Portfolio standard deviation = 13.17;
 Adjusted beta = 1.13.

MARKET TIMING IMPLICATIONS

Our study found a correlation between the number of stocks selling below liquidation value and the level of the market at that time. This was no surprise. The least squares regression graph (Chart A) re-



lating the number of stocks meeting the criteria of one of our portfolios with the level of the OTC index clearly illustrates this inverse relationship.

Although this is not a lagged relationship (useful as a leading indicator), the scarcity or availability of Graham's bargain issues appears to be a helpful guide to the technical position of the market. As a consequence, an individual investor following Graham's strict buying strategy should be insulated from more speculative market periods.

WHY DOES IT WORK?

Why do these simple statistical tests provide such superior results? The explanation may lie in a few of the points already examined.

First, large institutional investors can usually invest only in the top 1000 to 1500 high-capitalization stocks. With the advent of negotiated commission rates since 1975, stock research dollars and consequently the number of stocks closely followed by Wall Street research houses have been even further curtailed. This situation leads to a large second tier of stocks that may be inefficiently valued by the market.

We were unable to discover any "magic" qualities associated with stocks selling below liquidation value. Simply stated, by limiting our investments to stocks that according to fundamental notions of stock valuation appear *severely* depressed, we were able to locate more than our share of these inefficiently priced, undervalued securities. In other words, there are probably many more undervalued stocks that are *not* selling below liquidation value. Our statistical screens are merely exploiting a group of undervalued

stocks that are easily identified and are further protected by strong balance sheets and large asset values. Additionally, because of the depressed nature and liquid make-up of companies that meet our test criteria, they are often the object of takeover initiatives.

A more troubling question remains. If indeed so many inefficiently priced securities are available, why are so few people getting rich following Ben Graham's advice? As noted in the previous discussion, many of our stocks are too small for institutions to research or invest in. While a few professional groups currently manage modest sums in stocks meeting some of Graham's criteria, *our study would not have shown the results it did if the market for these stocks had already been exploited.*

Perhaps, as further computerization makes detailed financial information on second tier stocks more affordable and accessible to small investors, Graham's most easily identified opportunities will eventually disappear. Given the superior performance of subliquidation stocks for over 50 years, however, such a result may still not be imminent.

In any event, the greater significance to the enterprising investor should be the apparent opportunity to profit from fundamental research in the inefficient second tier. While systematic stock screening approaches using Graham's or other statistical criteria may not survive an era in which investors routinely review thousands of companies with their home com-

puters, those people who are willing to conduct more thorough analysis of secondary stocks should still be able to achieve impressive results.

Unless the structure and focus of the securities industry changes dramatically towards more coverage of secondary issues, a route made unlikely by basic economic realities, the individual may not have to worry about the prospect of a completely efficient second tier. On the other hand, in the more computerized future, it is likely that "bargain" stocks will become harder to find and will provide lower excess returns than the "bargain" stocks currently available. Even under these circumstances, the small investor should retain a significant advantage over the large institutional equity funds.

As Ben Graham noted in his classic, *The Intelligent Investor*, "It always seemed, and still seems, ridiculously simple to say that if one can acquire a diversified group of common stocks at a price less than the applicable net current assets alone — after deducting all prior claims, and counting as zero the fixed and other assets — the results should be quite satisfactory." Apparently, in today's world of efficient markets, investors can still profit from Ben Graham's sage advice.

¹ Data on nine additional portfolios are available on request from Joel Greenblatt, 5 Hickory Drive, Great Neck, N.Y. 11021.